



ECTE333: Microcontroller Architecture and Applications

Tutorial 2: Serial Communication

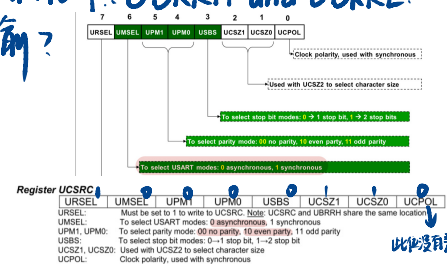
Related Lecture:

Lecture 3: Serial Communication

1. Serial communication in ATmega16: Quiz

- On what IO port do TXD and RXD pins reside? *TXD RXD在IO口的哪个位置(查附录)*
- What registers are used to set the baud rate? *什么寄存器用来设置波特率? UBRRH and UBRL.*
- How to select asynchronous transmission? *如何选择异步传输?*
- How to specify the number of stop bits?
- How to select parity bit options?
- How to double the transmission speed?
- How to specify the character size 5, 6, 8, or 9?

(XCK/TO) PB0	1	40	PA0 (ADC0)
(T1) PB1	2	39	PA1 (ADC1)
(INT2/AN0) PB2	3	38	PA2 (ADC2)
(OC2A/AN1) PB3	4	37	PA3 (ADC3)
(SCL) PB4	5	36	PA4 (ADC4)
(MOSI) PB5	6	35	PA5 (ADC5)
(MISO) PB6	7	34	PA6 (ADC6)
(SCK) PB7	8	33	PA7 (ADC7)
RESET	9	32	AREF
VCC	10	31	GND
GND	11	30	AVCC
XTAL2	12	29	PC0 (TOSC2)
XTAL1	13	28	PC0 (TOSC1)
(RXD) PDI	14	27	PC0 (TOD)
(TXD) PDI	15	26	PC0 (TOD)
(INT0) PDI	16	25	PC0 (TOD)
(INT1) PDI	17	24	PC0 (TOD)
(OC1A) PDI	18	23	PC0 (TOD)
(OC1B) PDI	19	22	PC0 (TOD)
(ICP1) PDI	20	21	PC0 (TOD)



2. Serial communication in ATmega16: Initialisation *初始化*

- Describe the steps and the relevant registers to initialise the serial port on the ATmega16 for asynchronous serial transmission. *异步串行通信初始化的步骤和寄存器*
- Write C code for the following asynchronous serial transmission settings *异步串行通信*
 - Baud rate 2400bps, 8 data bits, 1 start bit, 1 stop bit, no parity bit, system clock 1MHz. *开始 无校验位*
 - Baud rate 4800bps, 7 data bits, 1 start bit, 2 stop bits, even parity bit, system clock 4 MHz. *偶校验*

UCSZ2	UCSZ1	UCSZ0	Character Size
0	0	0	5-bit
0	0	1	6-bit
0	1	0	7-bit
0	1	1	8-bit
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Reserved
1	1	1	9-bit

3. Serial communication in ATmega16: Sending and receiving a character *收发字符*

- Explain the steps and the registers involved in sending a character through the serial port. Assume that the polling approach is used. *解释发送寄存器和流程图 发字符*
- Explain the steps and the registers involved in receiving a character through the serial port. Assume that the polling approach is used. *流程图 收字符*

4. Serial communication in ATmega16: Programming *编程*

Write a C program for the STK500 board that continually reads a character from the serial port of ATmega16 and sends the character to PORTA. Use transmission parameters in Question 2b-i.

5. Serial communication in ATmega16: Programming

A programmable pan-tilt camera MCAM100 is controlled through a serial connection: 8 data bit, 1 stop bit, no parity bit, baud rate 9600bps. The single character to be sent to the camera is given as

- '4' for left movement,
- '6' for right movement,
- '8' for up movement,
- '2' for down movement.

Write a C program for the STK500 board to repeatedly rotate the camera to the left 5 times, to the right 5 times, up 5 times and down 5 times.



6. Timing in asynchronous serial transmission

An asynchronous serial transmission scheme uses 8 data bits, 1 start bit, 1 stop bit and 1 even-parity bit. The incoming bits are always sampled at the middle of each clock period of the receiver. Determine the relationship between the clock frequency of the sender and the clock frequency of the receiver to prevent framing errors.

7. Asynchronous versus synchronous serial transmission

- Differentiate synchronous versus asynchronous serial transmission.
- Suppose a file of 10,000 bytes is to be sent over a line at 19600bps.
 - Calculate the overhead in bits and seconds if asynchronous communication is used. Assume that 1 start bit, 1 stop bit and 8 data bits are to be sent for each character, and no parity bit is used.
 - Calculate the overhead in bits and seconds if synchronous communication is used. Assume that the data are sent in frames, each 1000 characters and an overhead of 48 control bits per frame.

***** END OF TUTORIAL 2 *****